Why, then, are physicists constantly talking about quantum fluctuations? It's the same phenomenon we have noted in other contexts: we human beings have an irresistible urge to think of *what we see* as being real, even though quantum mechanics keeps telling us to do better. Hidden-variable theories give in to this urge by making something real other than the smoothly evolving wave function.

Everettian quantum mechanics is clear: empty space is described by a stationary, unchanging quantum state, where nothing is happening from moment to moment. But if we were to look sufficiently carefully, measuring the values of a quantum field in some small region, we would see what looked like a random mess. And if we looked again a moment later, we would see a different-looking random mess. The temptation to conclude that there is something moving around in empty space, even when we're not looking, is overwhelming. But that's not what's going on. Rather, we're seeing a manifestation of what we talked about in the context of the uncertainty principle: when we observe a quantum state, we typically see something quite different from what the state was before we looked.

To drive this point home, imagine that we do a more experimentally feasible measurement. Rather than measuring the value of a field at every point, let's just measure the total number of particles in the vacuum state of a quantum field theory. In an ideal thought-experiment world, we can imagine doing that measurement throughout all of space all at once. Since by construction we're in the lowest-energy state, you won't be surprised to hear that we will, with perfect confidence, detect no particles anywhere. It's just empty space. But in the real world, we will be confined to doing an experiment in some specific region of space, such as the interior of our laboratory, and asking how many particles there are. What should we expect to see?

This doesn't sound like a hard question. If there are no particles anywhere, then certainly we won't see any particles in our lab, right? Alas, no. That's not how quantum field theory works. Even in the vacuum state, if our experimental probe is confined to some finite region, there will always be a small probability of observing one or more particles. Generally the probability will be really, really small—not something we have to worry about in realistic experimental setups—but it will be there. The converse is also true: there will be quantum states for which our local experiment will